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Measurement of the optical properties of the skull in the wavelength range 650–950 nm

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Abstract. The optical properties of samples of bone from pig skull have been measured over the wavelength range 650–950 nm. The scattering phase function was measured on thin samples of the bone using a goniometer, and a value for the mean cosine g , of the scattering angle, was calculated. The scattering and absorption coefficients, μ_s and μ_a were then determined from measurements of diffuse reflectance and transmittance made with a pair of integrating spheres, by a step-wise search through a table of diffuse reflectance and transmittance versus μ_a and μ_s generated by a Monte Carlo model incorporating the measured scattering phase function. Values for g measured on six samples varied from 0.925 ± 0.014 at 650 nm to 0.945 ± 0.013 at 950 nm. Corresponding values for μ_a and μ_s measured on 18 samples were $\mu_a = 0.04 \pm 0.002 \text{ mm}^{-1}$, $\mu_s = 35 \pm 0.7 \text{ mm}^{-1}$ at 650 nm to $\mu_a = 0.05 \pm 0.002 \text{ mm}^{-1}$, $\mu_s = 24 \pm 0.6 \text{ mm}^{-1}$ at 950 nm.

1. Introduction

The use of near-infra-red spectroscopy (NIRS) to determine, non-invasively, the oxygenation of cerebral blood and tissue was first shown by Jöbsis (1977) in the cat. The technique has since been developed to monitor cerebral oxygenation in the human adult (Ferrari *et al* 1986, Elwell *et al* 1992) and infant (Brazy *et al* 1985, Wyatt *et al* 1986). There is now considerable interest in the development of NIRS imaging of oxygenation distribution in intact organs (Arridge *et al* 1986, 1991, Grünbaum *et al* 1991, Hebden *et al* 1990, Key *et al* 1991).

In order to quantify spectroscopy data and hence determine chromophore concentrations, it is necessary to know the pathlength travelled by the light through the tissues. This pathlength can be determined from measurements of the time of flight of ultrashort pulses of light through the head (Delpy *et al* 1988, Wyatt *et al* 1990, van der Zee *et al* 1992), or derived from a measurement of the phase shift of frequency modulated light (Lakowicz *et al* 1990, Sevick *et al* 1991). However, these measurements yield only the total path travelled by the light through all the tissues between the detector and source, and it is, at present, not possible to determine non-invasively the contribution of different regions to the overall pathlength.

In applications of NIRS to the brain, light must pass through the skull and surface tissue before entering or exiting the brain. In order to accurately model the effect of the tissue layers, it is necessary to know their optical properties, i.e. the scattering phase function and the scattering and absorption coefficients (μ_s, μ_a). The mean

cosine of scatter g , is also a useful parameter, for comparing results from Monte Carlo models to those employing diffusion theory, which use the reduced scattering coefficient, $\mu'_s = \mu_s(1 - g)$. Sevick and Chance (1991) did produce data from a two-layer lattice Monte Carlo model of the head, although they assumed a value of $\mu'_s = 6 \text{ mm}^{-1}$ for bone and 2 mm^{-1} as an average for brain, which is almost the opposite of accepted values for brain ($\sim 6 \text{ mm}^{-1}$) (Sternborg *et al* 1989, van der Zee 1993) and our value for bone ($\sim 2 \text{ mm}^{-1}$), and hence the results are of dubious value.

An extensive search of the literature has yielded no measured data on the optical properties of bone, the closest being data for the complex index of refraction of dentine at $10.0 \text{ }\mu\text{m}$, (Duplain *et al* 1987). We have therefore made measurements on samples of bone obtained from the adult pig skull. This bone was chosen for several reasons: Firstly, the physical and chemical composition of bone of the large mammals does not vary greatly (Hancox 1972) and hence pig bone is a good substitute for human bone. Secondly, the frontal/parietal bone of the pig is a flat plate of approximately 2 cm in thickness, from which it is easy to prepare samples of a suitable size.

2. Sample preparation

The heads of freshly slaughtered pigs were obtained from a commercial butcher, and the parietal bone was removed. All surface tissue was removed by careful scraping with a scalpel blade. The bone was then sawn into pieces approximately 20 mm square, the bone samples being about 15 mm thick. From these pieces, sections approximately 2 mm thick were cut, either parallel or perpendicular to the surface, using a slow-speed diamond saw (Buehler Ltd). These were then placed in a 0.9% saline solution in an ultrasonic water bath for 30 min to remove as much blood as possible. These samples were used for the measurement of diffuse reflectance and transmittance.

Thin bone slices for the scattering phase function measurement were prepared by grinding bone slices with 400 grit abrasive paper (Buehler Ltd), with the samples held in a water soluble wax (Crystalbond 555, Aremco Inc.). These were again washed in the ultrasonic bath to remove any grit. The sections prepared in this way were approximately 60–90 μm thick and 5 mm square.

3. Measurement procedure

3.1. Scattering phase function

The scattering phase function is the angular distribution of the light intensity due to a single scattering event. This was therefore measured using very thin samples ($\sim 60 \text{ }\mu\text{m}$, see discussion), to ensure that the incident light only scattered once. A collimated beam was used to illuminate the sample, and the scattered light measured around the sample with a collimated detector.

The system used in these measurements (figure 1(a)) was originally developed to measure the scattering phase function of brain tissue, and some details of the instrument have been described elsewhere (van der Zee and Delpy 1988). The system consists of a goniometer turntable, mounted onto which are a dual-pinhole collimator

coupled via an optical fibre to a broad-band light source (150 W halogen lamp, Oriel Ltd), and a collimated telescope connected via a similar optical fibre to a cooled charge coupled device (CCD) spectrophotometer (Cope *et al* 1989). The spectral range of the detector was 650–950 nm and the resolution, controlled by varying the light entrance slit width, was set at 10 nm. A PC compatible computer collected the spectra. To hold the sample, two glass hemi-cylinders were used, the sample being sandwiched between them to form a cylinder of 65 mm diameter. Ethylene glycol was used to provide optical coupling between the glass and bone, and the hemi-cylinders had an anti-reflection coating. The sample holder was mounted on the goniometer turntable, between the collimated light source and detector telescope. The cylinder was positioned so that the sample was oriented at 80° in the horizontal plane to the incident light beam direction. This reduced the errors arising from specularly reflected light which could otherwise be reflected at the cylinder circumference boundary and hence enter the detecting telescope.

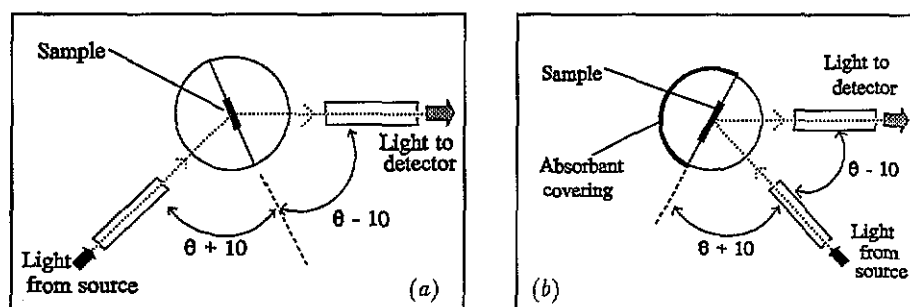


Figure 1. Goniometer arrangement for measuring phase function between (a) 0° and 90° and (b) 90° and 180° .

Measurements were made by scanning the incident beam through 170° , moving the sample at half this angular rate to ensure that the light path through the sample was kept to a minimum. When the source beam had been scanned through 90° , the cylinder was rotated through 90° (figure 1(b)), and measurements from this point were done in reflection, in order to avoid shallow angles of incidence of the light on the sample. A piece of black cloth impregnated with propylene glycol (refractive index 1.43) was attached to the back surface of the cylinder in order to reduce glass air reflections, which otherwise lead to a spurious rise in the phase function at angles approaching 180° .

The angular range over which measurements could be made was determined at low angles by the angular spread of the unscattered light beam, and above 170° by the physical dimensions of the collimating tubes. The angular response of the system was measured with ethylene glycol between the hemi-cylinders. The full width at half-maximum of the response was found to be 1.5° .

3.2. Absorption and scattering coefficients

To measure the absorption and scattering coefficients, the diffuse reflectance and transmittance of the sample resulting from collimated illumination were measured using two integrated spheres (figure 2).

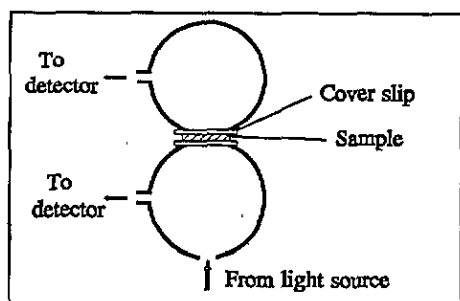


Figure 2. Integrating sphere system for measuring the diffuse transmittance and reflectance from collimated illumination of a sample slab.

The 2 mm thick sections of bone were placed between two glass cover slips, with saline providing optical coupling between glass and bone. The sample was then held between the 6 cm diameter integrating spheres. The cover slips were used in order to reduce diffusely reflected light from the surface of the bone. The sample was illuminated with a collimated light beam from a broad-band source and the diffuse light intensity in the two spheres measured with the CCD spectrophotometer, set to a resolution of 2 nm. The system was calibrated before each measurement using reflectance standards (Oriol Scientific).

4. Calculations

The two sets of data measured at 0–90° and 90–170° were fitted together by matching the values of scattered intensity at 90° for the reflection and transmittance measurements. As can be seen, this procedure results in a small peak in the phase function at 90° (figure 3). This peak is thought to be a measurement artifact due to the difference between transmission and reflection at 90° (see discussion). It is reasonable to assume that the phase function should be smoothly varying, and we therefore rejected the data between 80° and 100° and fitted a smoothly varying polynomial over this range, to extend the data to the full 180°. The mean cosine g , of the phase function, was calculated using

$$g = \frac{\sum f(\theta) \cos \theta \sin \theta}{\sum f(\theta) \sin \theta}$$

where $f(\theta)$ is the scattered light intensity as a function of angle, and azimuthal symmetry is assumed, the sine factor arising due to integration over the azimuthal angles.

Formulae to calculate exact diffuse reflectance and transmittance from integrating sphere measurements have been derived by Cheong (1990) and Pickering *et al* (1992). We used the formulae suggested by Cheong who considered the effect of losses through the portholes in the spheres, multiple traversal of light through the sample and the difference in reflection between the sample and sphere walls, and derived equations to correct for these effects, permitting true reflectance and transmittance to be calculated. These equations were further modified to correct for the effects of the reflection at the glass cover slips (van der Zee *et al* 1991). A Monte Carlo simulation of the experimental system was then run to generate a look-up table of transmittance and reflectance versus μ_a and μ_s . This simulation used the average of

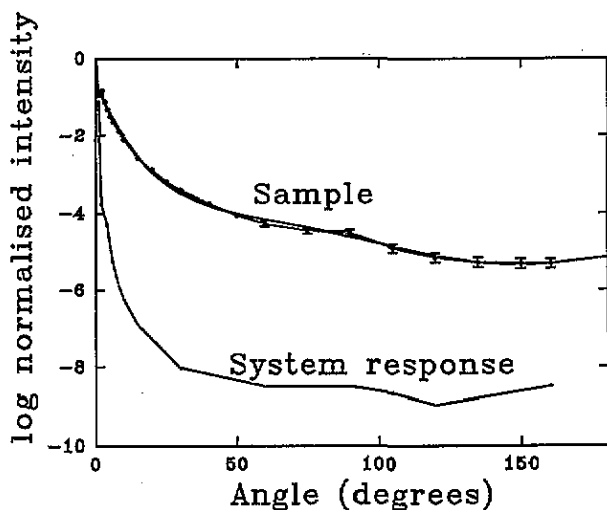


Figure 3. Average of the scattering phase function of skull at 800 nm from measurements on six samples, together with the system response of the goniometer, error bars ± 1 SD.

the experimentally derived scattering phase function for bone measured at 800 nm using the previously described goniometer system. The use of a single-wavelength scattering phase function was felt to be justified, since this varied very little with wavelength in the 650–950 nm range (see results). The measured reflectance and transmittance were then compared with the calculated values in the look-up table by a step-wise search using a Newton Raphson technique. Linear interpolation was then used between the closest data to give a more accurate estimate of μ_a and μ_s .

5. Results

Pieces of bone were obtained from a total of three pigs' heads. Six separate samples were taken from each for the integrating sphere measurements (i.e. 18 samples in total), and two samples from each were prepared for the phase function measurements (i.e. six samples in total). Figure 3 shows the average of the scattering phase functions measured at 800 nm, together with the angular response of the system. Figure 4 shows the reduced scattering coefficient, together with measured values of g over the wavelength range 650–950 nm. Finally, figure 5 shows the scattering coefficient together with the absorption coefficient.

6. Discussion

The scattering phase function is very highly forward peaked, having an average g value of 0.93, varying by only a few per cent over the wavelength range. The components of bone are hydroxyapatite (58%), collagen (25%), water (12%) and carbohydrate (5%) (White *et al* 1991). The average size of hydroxyapatite crystals is $30 \times 3 \times 3$ nm and that of collagen molecules $300 \times 1.5 \times 1.5$ nm (Posner 1987), which makes the particles Rayleigh scatterers. They should therefore have fairly isotropic scattering

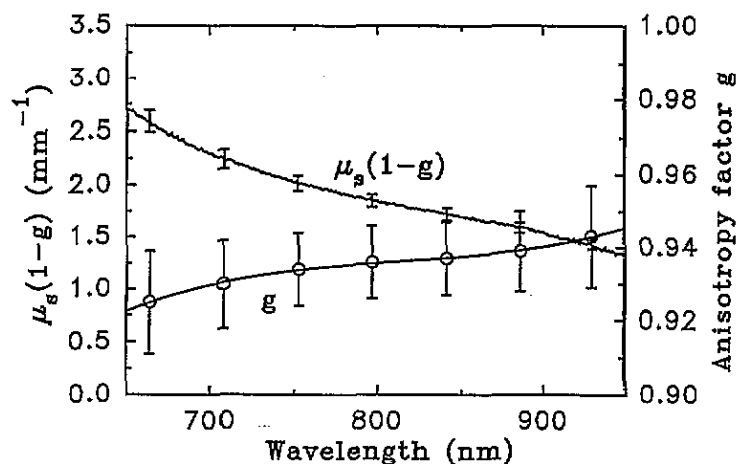


Figure 4. Average of the reduced scattering coefficient and anisotropy factor of skull from measurements on six samples, error bars ± 1 SD.

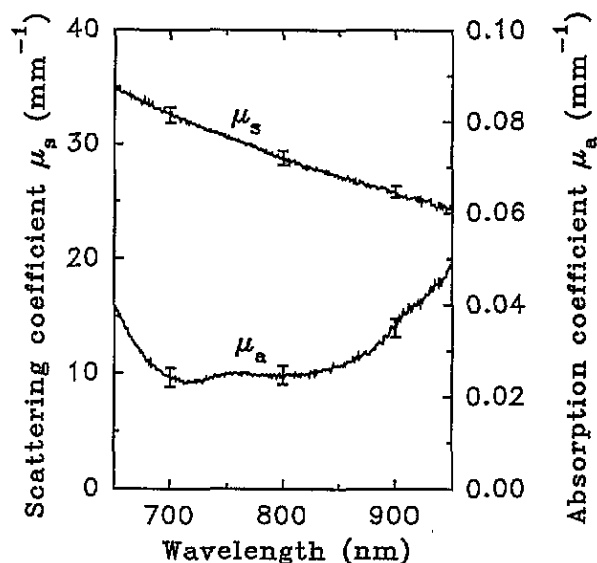


Figure 5. Average of the scattering and absorption coefficients measured on 18 samples of skull, error bars ± 1 SD.

properties. Since this is not the case, the scattering must therefore be dominated by conglomerations of the crystals which have a much greater scattering efficiency than the individual crystals.

In the goniometer measurements, the greatest source of error was due to reflections at boundaries between different media. Although this can be reduced by the use of ethylene glycol as a coupling medium, anti-reflection coatings etc, it cannot be removed totally. The scattering coefficient was measured as $\sim 30 \text{ mm}^{-1}$ which implies that the scattering length is $\sim 33 \mu\text{m}$. The samples used for scattering

phase function measurements were thicker than this, being 60–90 μm thick. It is difficult to prepare thinner samples, the hardness and porosity of the bone resulting in a tendency for the samples to crumble. This thickness will result in $\sim 50\%$ of the measured photons having undergone second/third-order scattering, which will tend to reduce the measured value of g . However, as the scattering phase function is so highly forward peaked, this will not make a great contribution, as most of the second/third-order scatters will be in the forward direction.

The scattering coefficient shows an almost linear fall with wavelength over this wavelength range, and there were no noticeable differences between data obtained from samples cut parallel or perpendicular to the skull surface.

The main features of the absorption spectra are the water peak above 900 nm, and a rise below 700 nm together with a small ripple at 750 nm, which are attributable to residual deoxygenated haemoglobin. Using a least-squares procedure to fit the known spectra of these components to the measured spectrum, it is possible to estimate that the water contributes $\sim 90\%$ of the absorption, and the haemoglobin 3–6%. Removing these from the measured spectra gives a low and largely featureless absorption spectrum for the solid bone.

For the integrating sphere measurements, the main source of error is in the Monte Carlo model. It is almost impossible to reproduce exactly the experimental conditions in a computer model. However, the model does provide a good approximation of reality, and the results should be accurate to within a few per cent.

As a result of the high g value, the reduced scattering coefficient of bone, surprisingly, is considerably less than that of brain white matter ($12\text{--}7.5\text{ mm}^{-1}$) and slightly less than that of brain grey matter ($4\text{--}2\text{ mm}^{-1}$) (Sternborg *et al* 1989, van der Zee 1992). It is also less than values quoted for both dermis and epidermis ($\sim 4\text{ mm}^{-1}$), (van Gemert *et al* 1989). This would imply that in NIRS studies on the adult head, the effect of light scattering by the skull is of the same order of magnitude as that of the surrounding layers of skin and brain. Knowledge of the optical properties of bone will enable Monte Carlo modelling of light transport in the head, incorporating the different layers of tissue, to be carried out, with the aim of determining the relative contributions of each tissue layer to the detected signal.

Acknowledgments

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